

EXPERIMENT NO: 19

BLINKING LEDs USING SOFTWARE DELAY ROUTINE IN LPC2148 KIT

AIM

To write and execute a C program to blink LEDs using software delay routine in LPC2148 kit.

OBJECTIVES

- Configure Port 0 of the LPC2148 as a general-purpose output port.
- Turn the LEDs ON and OFF using the IOSET0 and IOCLR0 registers.
- Implement a software delay routine to obtain visible LED blinking.

APPARATUS / SOFTWARE REQUIRED

- Keil uVision5 Software
- Philips Flash Programmer
- LPC 2148 kit

PROGRAM

```
#include "lpc214x.h"

void delay (unsigned int k);

void main(void)
{
    IODIR0 = 0xFFFFFFFF;    //Configure Port0 as output Port
    PINSEL0 = 0;            //Configure Port0 as General Purpose IO

    while(1)
    {
        IOSET0 = 0x0000ff00;    //Set P0.15-P0.8 to '1'
        delay(1000);            //1 sec Delay
        IOCLR0 = 0x0000ff00;    //Set P0.15-P0.8 to '0'
        delay(1000);            //1 Sec Delay
    }

    //Delay Program
    //Input - delay value in milli seconds
    void delay(unsigned int k)
    {
        unsigned int i,j;
        for (j=0; j<k; j++)
            for(i = 0; i<=800; i++);
    }
}
```

OUTPUT

LEDs P0.15-P0.8 are blinking

RESULT

Thus the C program to blink LEDs using software delay routine was written and executed in LPC 2148 kit.